

IntentVC 2025: The ACM Multimedia Grand Challenge on Intention-Oriented Controllable Video Captioning

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Abstract

The IntentVC Challenge, held in conjunction with ACM Multimedia 2025, introduces a novel benchmark for intention-oriented controllable video captioning. Unlike conventional captioning methods that generate generic, scene-level summaries, IntentVC focuses on intention-specific generation. Participants are required to produce captions explicitly conditioned on user-defined intentions, such as emphasizing a specific object tracked within a video. To support this task, the challenge provides an extended version of the LaSOT dataset annotated with intention-focused captions across 70 object categories. A standardized evaluation protocol and public leaderboard enable fair and reproducible comparison among submitted methods. By advancing research in personalized and adaptive video understanding, IntentVC offers a platform for exploring controllable vision-language modeling with practical relevance for accessibility, retrieval, and human-AI interaction. As a result, a total of 23 teams and 58 active participants have participated, and a total of 1,443 entries have been submitted. More information and resources are available at <https://sites.google.com/view/intentvc/>.

CCS Concepts

• Computing methodologies → Computer vision tasks.

Keywords

Video Captioning, Controllable Captioning, Intention-Oriented AI, Vision-Language Models, Multimedia Understanding

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1 Introduction

Video captioning [1] has emerged as a fundamental task in multimedia understanding, enabling systems to generate natural language descriptions of dynamic visual content. While recent advances have significantly improved the fluency and contextual relevance of generated captions, existing approaches largely produce generic, one-size-fits-all narratives. Existing models often fail to account for user-specific focus or intent, which is critical in scenarios where viewers seek information about particular objects, actions, or contextual cues within a video.

This limitation poses a significant barrier to deploying captioning systems in real-world applications such as assistive technology, personalized content delivery, and intention-driven media retrieval. To bridge this gap, the IntentVC Challenge introduces a novel benchmark for *intention-oriented controllable video captioning*. The primary objective is to promote the development of models that can condition caption generation on user-defined intentions, enabling adaptive and personalized video understanding. By supporting fine-grained control over the captioning process, IntentVC aims to catalyze research in user-aligned vision-language modeling and expand the boundaries of interactive multimedia systems.

2 Challenge Overview

The first edition of the IntentVC challenge centers on the task of controllable video captioning conditioned on object-level intention cues. In this challenge, participants are provided with a video and an associated object tracklet, defined as a temporally ordered sequence of bounding boxes that identify a specific object of interest throughout the video. The objective is to generate a contextually appropriate caption that emphasizes the visual narrative of the specified object, while maintaining relevance to the overall scene.

To support this task, the challenge introduces an extended version of the LaSOT dataset [2], a widely adopted benchmark originally developed for large-scale single-object tracking. For IntentVC, LaSOT has been enriched with intention-oriented caption annotations across 70 object categories. Each annotated caption is crafted to reflect the perspective of the target object as defined by its tracklet, providing a grounded textual description aligned with the user-specified focus. This setup enables fine-grained evaluation of a model's ability to adapt caption content in response to explicit object-level conditioning.

3 Challenge Evaluation

To ensure rigorous and standardized evaluation, all submissions are evaluated using widely adopted metrics in video captioning. These metrics measure linguistic accuracy, and semantic alignment with the reference captions. Specifically, the following metrics are used:

- **BLEU@4**: Measures four-gram precision to assess local fluency and surface-level match.
- **METEOR**: Computes a weighted F-score using exact, stem, synonym, and paraphrase matches for robust scoring.
- **CIDEr**: Uses TF-IDF to evaluate consensus with reference captions, capturing sentence-level semantics.
- **ROUGE-L**: Measures the longest common subsequence to reflect structural and sequential alignment.

Together, these metrics provide a comprehensive assessment of both surface-level and semantic caption quality, enabling reproducible comparisons across systems.

4 Challenge Results

The IntentVC challenge garnered substantial interest from the multimedia research community, particularly in the fields of video understanding and vision-language modeling. A total of 23 teams officially registered for the challenge, with 58 active participants engaging on the submission platform. Throughout the competition period, participants submitted a total of 1,443 entries, demonstrating strong engagement and continuous model refinement. This level of activity highlights the growing interest in intention-oriented controllable video captioning and its potential to drive advancements in multimedia systems. The competition was highly competitive, and the narrow performance margins among leading entries underscore the rigor of the task and establish IntentVC as a strong benchmark for controllable video captioning (see Table 1).

The top-performing teams proposed diverse and effective approaches for intention-driven video captioning.

- **Rank 1 – MGVC (clever_knight)**: A two-stage framework leveraging multimodal large language models (MLLMs). A fine-tuned MLLM generates diverse candidate captions, which are refined by a second MLLM for better alignment and fluency. A video-text matching module filters misaligned outputs, while category-specific regressors score and select the best caption. This method achieves high controllability, grounding, and stylistic quality.
- **Rank 2 – IntentVCNet (ustc-iai)**: Addresses spatio-temporal gaps in LVLMs using a prompt combination strategy and a novel box adapter. The adapter injects object-level semantics into the visual stream, while the prompt strategy links user intent with video segments. These enhancements improve fine-grained spatial control and temporal reasoning in caption generation.
- **Rank 3 – CMA-VC (ustc-iat-united)**: A unified cross-modal captioning pipeline based on InternViT and InternLM. The approach employs Low-Rank Adaptation for parameter-efficient fine-tuning. Visual content and user intentions are processed jointly, with spatial and temporal features dynamically aligned to generate accurate, intention-focused captions.

Table 1: Final results of all participating teams in the IntentVC Challenge. The best and second-best scores for each metric are highlighted in bold and underlined, respectively.

Team Name	BLEU@4 ↑	METEOR ↑	CIDEr ↑	ROUGE-L ↑
clever_knight	53.08	65.96	248.58	64.52
ustc-iai	<u>45.13</u>	<u>62.46</u>	<u>222.25</u>	<u>60.84</u>
ustc-iat-united	42.30	61.93	219.24	59.45
flyfish	43.20	61.05	214.90	59.67
dejie	42.12	60.49	206.11	59.08
triumph	41.16	60.60	210.26	58.71
neo_scene	40.20	60.14	207.15	58.26
gjc0714	36.53	56.87	184.96	55.59
ferry_li30	37.20	55.14	185.54	54.26
tamako0123	34.68	55.00	178.11	53.62
codzyong812	34.87	54.93	179.59	53.98
jszrz	33.99	54.56	175.07	53.17
zzyyxxznb	35.86	59.06	175.45	57.46
aaxia	33.17	55.83	176.52	53.59
datacatsam	30.42	53.54	163.64	51.13
yyby_zzxp	22.14	43.48	115.33	44.35
reisen	15.81	38.82	101.71	39.78
maiphuong	10.27	35.50	81.43	36.17
amitjaiswal	9.85	34.52	54.87	34.37
mousi30	6.84	28.59	37.63	33.40
ko4ro	2.48	26.51	14.10	23.17

These methodologies underscore the importance of combining MLLM tuning, spatial-temporal grounding, and intent modeling for controllable video captioning.

5 Conclusion

The IntentVC challenge introduced a new direction for video captioning by focusing on intention-oriented controllability. By providing a high-quality, intention-annotated dataset and establishing a clear evaluation framework, the challenge created a benchmark for developing and comparing models that generate captions aligned with user-specified focus. The dataset, evaluation platform, and supporting resources will remain publicly available to encourage further research in personalized, adaptive video understanding and to advance the capabilities of multimedia AI systems. In future editions, we will explore a broader range of objects and introduce more flexible conditioning by providing only partial bounding box information. This aims to better reflect real-world scenarios and encourage models to reason more effectively about user intent.

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